CSE 472 Assignment 2: Decision Trees, Random Forests, and Extra Trees

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1. Introduction

This report presents the implementation and analysis of tree-based classification algorithms from scratch, including Decision Trees, Random Forests, and Extremely Randomized Trees (Extra Trees). The implementations are compared against scikit-learn's optimized versions on the Iris and Wine datasets.

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2. Algorithmic Details

2.1 Decision Tree

Splitting Criterion: Gini Impurity

$$Gini(D) = 1 - \sum_{i=1}^{c} p_i^2$$

where p_i is the proportion of samples belonging to class i.

Information Gain:

$$IG(D, A) = Gini(D) - \sum_{v \in Values(A)} \frac{|D_v|}{|D|} Gini(D_v)$$

Key Implementation Details:

- Recursively builds tree by finding best split at each node
- Stores class distribution in leaf nodes for soft probability estimates
- Configurable: `max_depth`, `min_samples_split`, `max_features`

2.2 Random Forest

Algorithm:

1. For each tree $t = 1, ..., n\_estimators$:
    - Create bootstrap sample D_t from training data (sampling with replacement)
    - At each node, consider only $\sqrt{n\_features}$ random features

- Build decision tree on D_t

2. Prediction: Majority vote across all trees

Key Differences from Single Tree:

- Bootstrap sampling reduces variance through averaging
- Feature randomization decorrelates trees

2.3 Extra Trees (Extremely Randomized Trees)

Algorithm:

1. For each tree $t = 1, ..., n\_estimators$:
   - Use entire training set (NO bootstrap)
   - At each split: randomly select threshold for each candidate feature
   - Select split with best information gain among random thresholds
2. Prediction: Majority vote across all trees

Key Differences from Random Forest:

- No bootstrap → uses full dataset
- Random thresholds → faster training, potentially more variance reduction

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3. Experimental Setup

3.1 Datasets

Dataset	Samples	Features	Classes
Iris	150	4	3
Wine	178	13	3

3.2 Hyperparameters

Parameter	Value
`max_depth`	10
`min_samples_split`	2
`n_estimators`	100
`max_features`	sqrt
`random_state`	42

3.3 Evaluation Protocol

- **5-Fold Stratified Cross-Validation** for reliable performance estimates
- Results reported as Mean ± Standard Deviation

3.4 Metrics

1. **Accuracy:** Proportion of correct predictions
2. **F1-Score (Macro):** Harmonic mean of precision and recall, averaged across classes
3. **AUROC:** Area Under ROC Curve (One-vs-Rest for multi-class)

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4. Results

4.1 Iris Dataset (5-Fold Cross-Validation)

Model	Accuracy	F1-Macro	AUROC
Custom Decision Tree	0.9533	0.9531	0.9650
Custom Random Forest	0.9467	0.9464	0.9947
Custom Extra Trees	0.9600	0.9598	0.9937
Sklearn Decision Tree	0.9533	0.9531	0.9650
Sklearn Random Forest	0.9533	0.9531	0.9947
Sklearn Extra Trees	0.9600	0.9599	0.9943

4.2 Wine Dataset (5-Fold Cross-Validation)

Model	Accuracy	F1-Macro	AUROC
Custom Decision Tree	0.8937	0.8951	0.9185
Custom Random Forest	0.9773	0.9779	0.9972
Custom Extra Trees	0.9887	0.9892	0.9990
Sklearn Decision Tree	0.8881	0.8887	0.9137
Sklearn Random Forest	0.9832	0.9836	0.9992
Sklearn Extra Trees	0.9775	0.9774	0.9992

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5. Analysis

5.1 Custom vs Scikit-learn Comparison

Algorithm	Dataset	Accuracy Δ	F1-Macro Δ
Decision Tree	Iris	+0.00%	+0.00%
Decision Tree	Wine	+0.56%	+0.64%
Random Forest	Iris	-0.67%	-0.67%
Random Forest	Wine	-0.59%	-0.57%
Extra Trees	Iris	+0.00%	-0.01%

Algorithm	Dataset	Accuracy Δ	F1-Macro Δ
Extra Trees	Wine	+1.13%	+1.18%

Observation: Custom implementations achieve comparable performance to scikit-learn, with differences typically within ±1%.

5.2 Ensemble Methods vs Decision Tree

Wine Dataset - Improvement over Decision Tree:

- Random Forest: **+8.37%** accuracy
- Extra Trees: **+9.51%** accuracy

5.3 Bias-Variance Analysis

Aspect	Single Decision Tree	Ensemble Methods
Bias	Low	Low
Variance	High (std: 0.054-0.064)	Low (std: 0.014-0.033)
Overfitting	Prone	Resistant

Key Findings:

1. Single decision trees show higher variance (larger standard deviations)
2. Ensemble methods significantly reduce variance through averaging
3. Random Forest uses bootstrap to create diversity; Extra Trees uses random thresholds
4. Both ensemble methods achieve near-perfect AUROC (>0.99) on Wine dataset

5.4 Random Forest vs Extra Trees

Characteristic	Random Forest	Extra Trees
Data per tree	Bootstrap sample	Full dataset
Split selection	Optimal threshold	Random threshold
Training speed	Slower	Faster
Variance	Low	Potentially lower

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6. Conclusion

This assignment successfully implemented Decision Trees, Random Forests, and Extra Trees classifiers from scratch. Key achievements:

1. **Correct Implementation:** All three algorithms produce results comparable to scikit-learn
2. **Bias-Variance Tradeoff:** Empirically demonstrated that ensemble methods reduce variance
3. **Configurable Hyperparameters:** All required parameters (max_depth, min_samples_split, n_estimators, max_features) are configurable
4. **Robust Evaluation:** 5-fold cross-validation provides reliable performance estimates

The ensemble methods consistently outperform single decision trees, validating the theoretical understanding that combining multiple weak learners reduces overfitting and improves generalization.